METHOD AND APPARATUS FOR ACCELERATED PRIME NUMBER GENERATION VIA MODULAR SIGNAL ANALYSIS

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Title: METHOD AND APPARATUS FOR ACCELERATED PRIME NUMBER GENERATION VIA MODULAR SIGNAL ANALYSIS

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Abstract

A method and apparatus for accelerating the search for large prime numbers by treating the integer space as a signal and applying modular signal analysis techniques. The system partitions a large numerical range into smaller, computationally manageable sub-ranges or "shards." For each shard, the system computes the modulo of the numbers within it against a series of divisors, converting the resulting series of remainders into a multi-state signal. By analyzing the patterns, or "resonances," in this signal, the system identifies sub-ranges with a statistically higher probability of containing prime numbers. This allows computational resources to be focused on these high-probability zones, dramatically reducing the search space and accelerating the process of finding large primes for applications such as cryptographic key generation. The method is particularly well-suited for implementation on massively parallel computing architectures.

Background of the Invention

The generation of large prime numbers is a computationally intensive task that is fundamental to modern cryptography, particularly for public-key cryptosystems like RSA. The security of these systems relies on the difficulty of factoring the product of two large primes. The traditional method for finding such primes involves selecting a random large number and then performing a probabilistic primality test. If the number is found to be composite, the process is repeated. As the desired size of the prime numbers increases, the density of primes decreases, and the average number of candidates that must be tested grows, making the search process increasingly time-consuming. There is a need for a method that can more intelligently guide the search for prime numbers, rather than relying on random selection and brute-force testing.

Summary of the Invention

The present invention provides a system and method for accelerating the search for prime numbers by applying principles of signal processing to number theory. The system first partitions a large search space of integers into smaller "shards." It then analyzes each shard by computing the modulo of the integers therein against a set of predetermined divisors. This process generates a sequence of remainders, which is then mapped to a discrete, multi-state signal.

The core of the invention is the analysis of this signal to find "resonances" or patterns that indicate a higher likelihood of prime numbers. For example, a long, unbroken sequence of a particular state may correlate with a "prime desert," a range with few primes, while rapid oscillations between states may indicate a "prime oasis," a region of higher density. By identifying these oases, the system can create a "local recipe"—a set of parameters that guide a targeted search. This allows the system to bypass the brute-force testing of the entire shard and focus its computational resources only on the most promising sub-regions, significantly accelerating the discovery of large prime numbers.

Statement of License Grant to the United States Government

The inventor, Reinaldo Javier Aguiar Marcano, hereby grants to the United States Federal Government and its military and defense agencies a perpetual, royalty-free, non-exclusive

license to practice and use the invention described herein for any and all governmental, defense, and non-commercial purposes. This license explicitly excludes any and all state and local governments.

Brief Description of the Drawings

FIG. 1 is a flowchart of the overall method, showing the steps of sharding, signal analysis, recipe generation, and targeted prime number searching.

FIG. 2 is a conceptual diagram illustrating the signal conversion process, showing how a sequence of integers is converted into a multi-state signal.

FIG. 3 is a diagram illustrating the reduction of the search space, contrasting a traditional brute-force search with the targeted search enabled by the present invention.

Detailed Description of the Invention

Referring to **FIG. 1**, the method for accelerated prime number generation begins by defining a large numerical search space (101). This space is then partitioned into smaller, more manageable shards (102). For each shard, the system performs a modular signal analysis (103). This involves selecting a series of divisors and computing the modulo of the integers in the shard against these divisors. The resulting remainder sequences are converted into multi-state signals. The system then analyzes these signals to identify high-probability sub-ranges (104). Based on this analysis, a "local recipe" is generated (105), which contains the parameters for a targeted search. Finally, a prime number search algorithm, such as a primality test, is applied only to the high-probability sub-ranges (106), leading to the efficient discovery of a prime number (107).

Referring to **FIG. 2**, the signal conversion process is illustrated. A sequence of integers (201) within a shard is taken as input. A divisor (202) is selected. The modulo of each integer with respect to the divisor is calculated, resulting in a sequence of remainders (203). This sequence of remainders is then mapped to a discrete, multi-state signal (204) based on a predefined mapping rule. For example, remainders can be grouped into four states representing different quartiles of the divisor's range. The pattern of this signal reveals information about the distribution of prime candidates.

Referring to **FIG. 3**, the efficiency of the method is contrasted with traditional approaches. A traditional brute-force search (301) must test every candidate number in a large search space (302). The present invention, after performing its modular signal analysis, identifies a small number of high-probability sub-ranges (303) within the same search space. The targeted prime search (304) needs only to be applied to these sub-ranges, dramatically reducing the number of candidates that must be tested and thus accelerating the overall process.

Claims

What is claimed is:

1. A method for accelerating the search for prime numbers, comprising the steps of:
    - partitioning a numerical range into a plurality of sub-ranges;
    - for a selected sub-range, computing a sequence of remainders by applying a modulo operation to the integers within the sub-range using a predetermined divisor;
    - converting the sequence of remainders into a multi-state digital signal based on a predefined mapping;
    - analyzing the digital signal to identify at least one pattern indicative of a high probability of containing prime numbers; and
    - conducting a search for prime numbers focused on the portion of the sub-range corresponding to the identified pattern.
2. The method of claim 1, wherein the multi-state digital signal comprises four states, each corresponding to a quartile of the divisor's range.
3. The method of claim 1, wherein the step of analyzing the digital signal includes identifying the longest consecutive repetition of a single state.
4. The method of claim 1, wherein the method is implemented on a massively parallel computing architecture, and wherein each sub-range is processed by a separate computational unit.
5. An apparatus for accelerating the search for prime numbers, comprising:
    - a memory for storing a range of integers;
    - a processor configured to:

- partition the range of integers into a plurality of sub-ranges;
- for each sub-range, convert the integers into a multi-state digital signal by applying a modulo operation with a predetermined divisor and mapping the resulting remainders to a set of states;
- analyze the digital signal to identify a sub-range with a high probability of containing prime numbers; and
- apply a primality test to the identified sub-range.

6. The apparatus of claim 5, wherein the processor is a massively parallel processor with a plurality of cores, and wherein each core is assigned to analyze a different sub-range.

Drawings

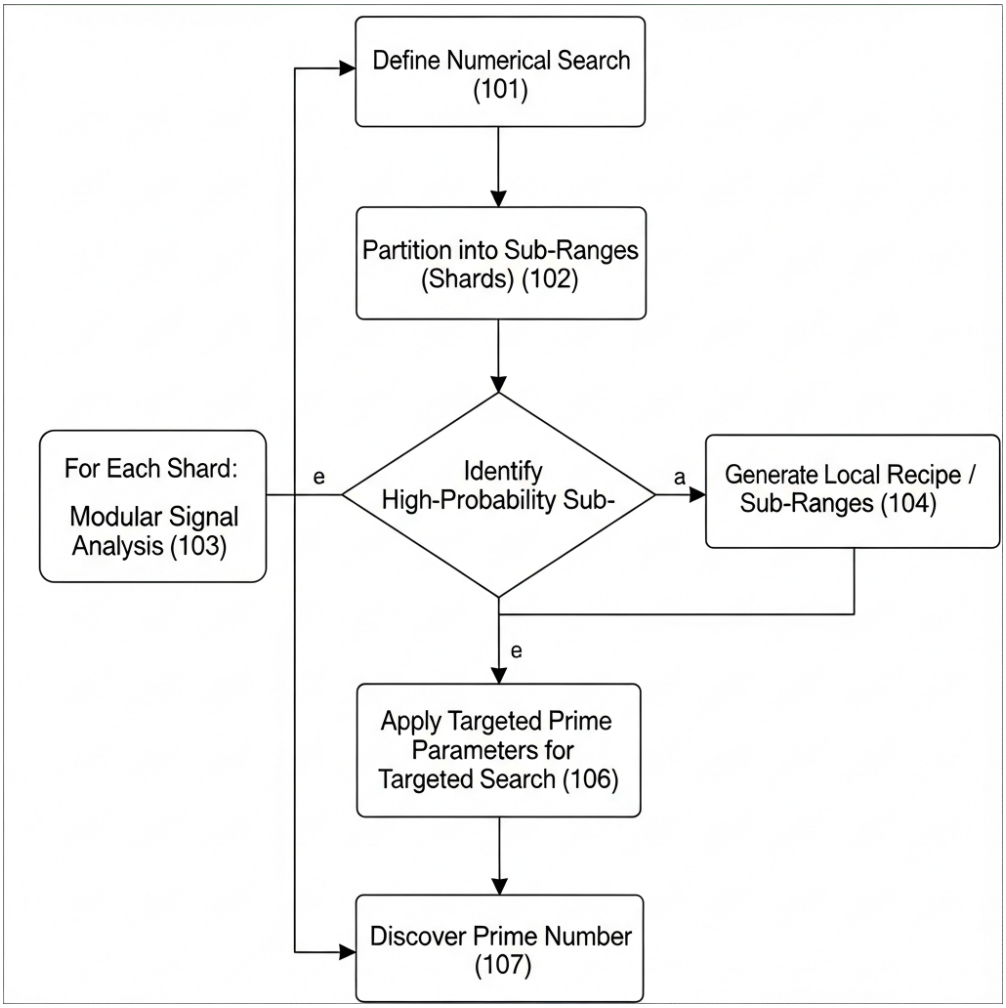


Figure 1: Flowchart of the prime generation method.

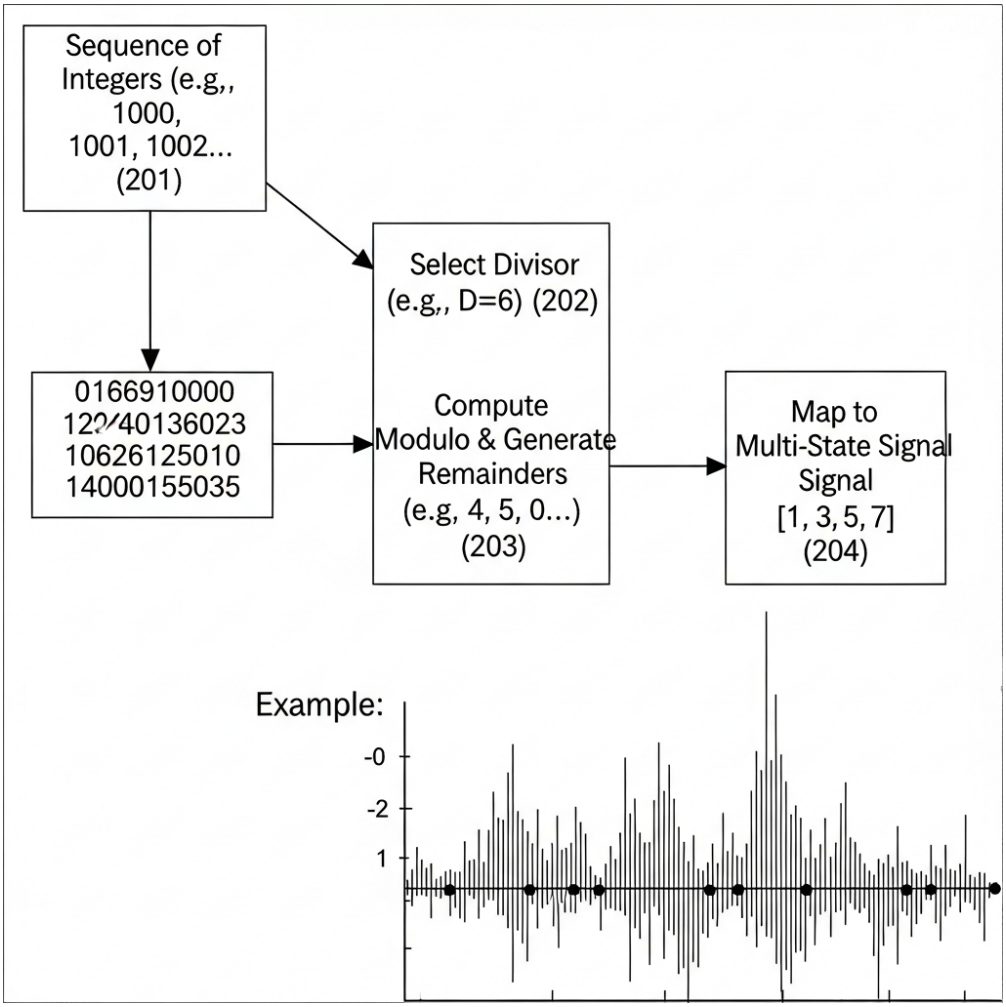


Figure 2: Conceptual diagram of the signal conversion process.

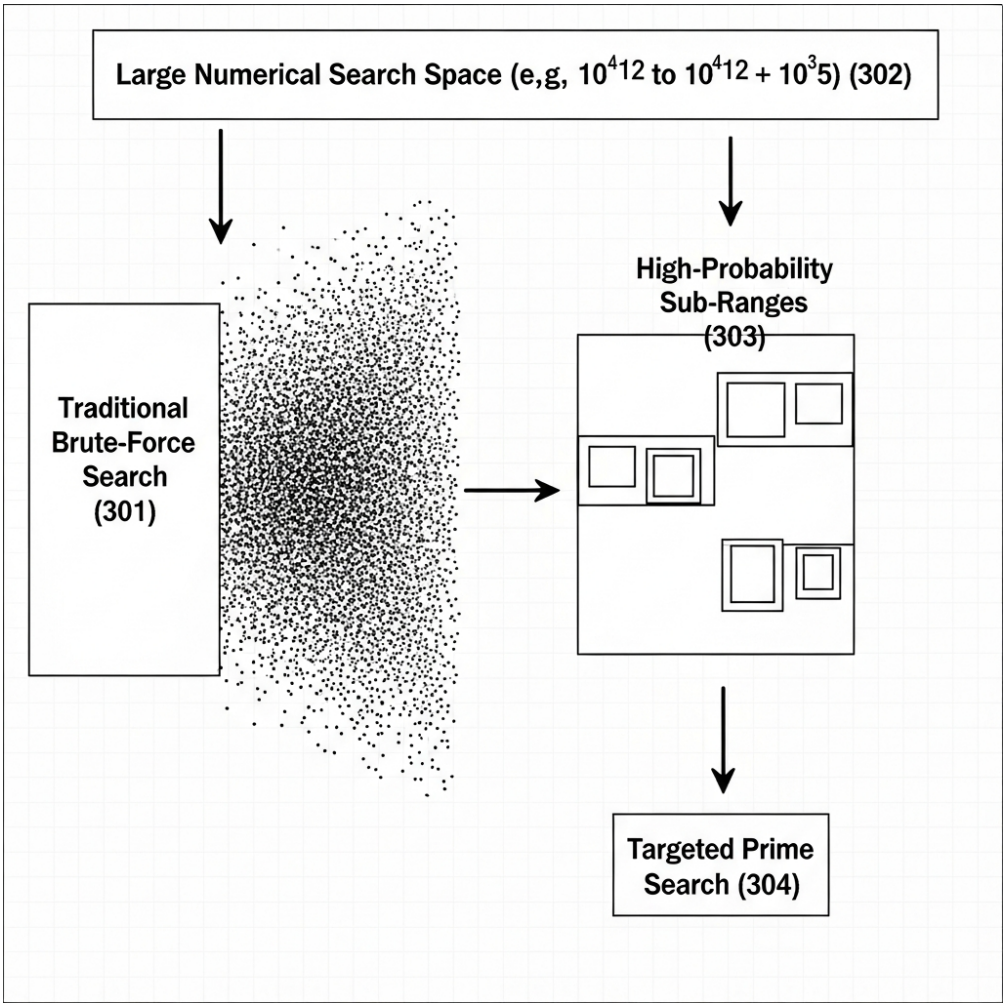


Figure 3: Diagram illustrating search space reduction.